

The empirical recurrence for OEIS A298550, recovered from its tail

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Abstract

OEIS A298550 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = -240 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A298550 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 2, 3, 5 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

$$1, 18, 14, 33, 74, 226, 716, 2211, \dots$$

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jan 21 2018 06:45:44, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	5	c_{19}	5583	c_{37}	−1055682	c_{55}	129366
c_2	−4	c_{20}	−7777	c_{38}	12961	c_{56}	118724
c_3	−2	c_{21}	−19708	c_{39}	1190260	c_{57}	−144075
c_4	−19	c_{22}	12921	c_{40}	−310079	c_{58}	−299957
c_5	18	c_{23}	45144	c_{41}	−437609	c_{59}	−78791
c_6	−3	c_{24}	27890	c_{42}	776336	c_{60}	102903
c_7	73	c_{25}	−40602	c_{43}	1416685	c_{61}	84185
c_8	−152	c_{26}	−39974	c_{44}	833651	c_{62}	−8761
c_9	316	c_{27}	−118542	c_{45}	−1184695	c_{63}	42782
c_{10}	124	c_{28}	−130441	c_{46}	59566	c_{64}	84615
c_{11}	136	c_{29}	−128441	c_{47}	−421289	c_{65}	−63910
c_{12}	−1458	c_{30}	278527	c_{48}	−753017	c_{66}	−41986
c_{13}	−883	c_{31}	323849	c_{49}	−1443639	c_{67}	28336
c_{14}	507	c_{32}	119083	c_{50}	115463	c_{68}	−5532
c_{15}	1783	c_{33}	345372	c_{51}	1119193	c_{69}	−7700
c_{16}	−261	c_{34}	342332	c_{52}	1230495	c_{70}	4672
c_{17}	−1174	c_{35}	31822	c_{53}	−117278	c_{71}	632
c_{18}	5396	c_{36}	−1460796	c_{54}	−567228	c_{72}	−240

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A298550 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = 240$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -240 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A298550.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.